

## Exercise Sheet

### Task 1 – Decimal to Binary Conversion

Convert the following decimal numbers into their binary representations. Make sure that you use enough bits to display the sign correctly.

a)  $(125)_{10}$

$$(125)_{10} = (01111101)_2$$

b)  $(39)_{10}$

$$(39)_{10} = (0100111)_2$$

### Task 2 – Binary to Decimal Conversion

Convert the following unsigned binary numbers into their decimal representations by using the specified conversion method.

a)  $(11101)_2$ , conversion table

|            |           |           |           |           |
|------------|-----------|-----------|-----------|-----------|
| $2^4 = 16$ | $2^3 = 8$ | $2^2 = 4$ | $2^1 = 2$ | $2^0 = 1$ |
| 1          | 1         | 1         | 0         | 1         |

$$16 + 8 + 4 + 1 = 29$$

b)  $(10101001)_2$ , multiply and add

$$\begin{aligned} 0 + 1 &= 0 \\ 1 * 2 + 0 &= 2 \\ 2 * 2 + 1 &= 5 \\ 5 * 2 + 0 &= 10 \\ 10 * 2 + 1 &= 21 \\ 21 * 2 + 0 &= 42 \\ 42 * 2 + 0 &= 84 \\ 84 * 2 + 1 &= 169 \end{aligned}$$

**Task 3** – More Conversion

Convert the following numbers into their representation from the specified number system.

- a)  $(1011101)_2$  to the octal system

$(135)_8$

- b)  $(754)_8$  to the hexadecimal system

$(1EC)_{16}$

**Task 4** – Two's Complement

Convert the following numbers into their two's complement representation. Use 8 bit to display the negative numbers.

- a)  $(-42)_{10}$

Value:  $(00101010)_2 \rightarrow$  OC:  $(11010101)_2 \rightarrow$  TC:  $(11010110)_2$

- b)  $(-15)_{10}$

Value:  $(00001111)_2 \rightarrow$  OC:  $(11110000)_2 \rightarrow$  TC:  $(11110001)_2$

**Task 5** – Sign and Zero Extension

Use sign and zero extensions appropriately to put these 5-bit numbers into an 8-bit format.

- a)  $(01010)_2$  signed

$(00001010)_2$

- b)  $(11010)_2$  signed

$(11111010)_2$

c)  $(01010)_2$  unsigned

$(00001010)_2$

d)  $(11010)_2$  unsigned

$(00011010)_2$

### Task 6 – Binary Operations

Compute the following operations.

a)  $(011010)_2 + (0101101)_2$

$(1000111)_2$

b)  $(1010)_2 * (1101)_2$

$(10000010)_2$

c)  $(110110)_2 XOR (101101)_2$

$(011011)_2$

d)  $(1011010)_2 AND (1101110)_2$

$(1001010)_2$